ChatGPT

Investigating optimal model architecture for Estonian day-ahead electricity price forecasting

Market setting and design principles

The forecasting problem is unusually constrained. The target is the day-ahead price path for Estonia [1] in the coupled auction operated through Nord Pool [2] and the pan-European Single Day-Ahead Coupling run with the EUPHEMIA algorithm. At 10:00 CET, available interconnector and grid capacities are published; final bids close at 12:00 CET; the auction then clears simultaneously across bidding zones. That means any feature derived from D+1 neighbor-zone clearing prices is illegitimate at forecast time, because those prices are set in the same simultaneous optimization. Only pre-noon fundamentals, lagged prices, published capacities, outages, weather forecasts, and other ex ante information are allowed. [3]

The dataset is also structurally difficult. Elering [4] states that the Baltic systems synchronized with Continental Europe on 9 February 2025, creating a real market-structure break, not just a statistical drift. Separately, European electricity prices were heavily distorted during the gas-crisis period, and recent probabilistic EPF work explicitly treats the post-2021 and post-2022 period as a regime shift that challenged established models. On top of that, the day-ahead market moved to 15-minute MTUs in SDAC on 30 September 2025, while the Baltic intraday 15-minute rollout started earlier, adding target-definition fragmentation even inside the “usable” year. [5]

That combination drives the architecture choice more than any fashionable model class. With only about 12 months of in-regime history, the true sample size for a daily 24-output model is roughly 365 target days, not 8,760 hourly points in the usual i.i.d. sense. The winning design should therefore prioritize three things: aggressive regularization, borrowing strength from related markets or structural features, and daily recalibration. In this setting, the main risk is not underfitting but learning unstable, non-transportable patterns from one annual cycle and a handful of extreme events. This is strongly aligned with the EPF best-practice literature, which emphasizes robust benchmarks, rolling recalibration, and significance testing over architectural novelty. [6]

A further design implication is that the target itself must be specified carefully. Official day-ahead market rules define the delivery day in CET/CEST market time, not local Estonian civil time. Since the post-September 2025 native SDAC object is quarter-hourly, a “24 hourly clearing prices” problem is no longer fully native to the market. For production, I recommend a two-layer definition: store the raw market-time native series as published, but build the initial forecasting product on a stable hourly decision layer that is explicitly defined and reproducible. If this is not fixed now, model comparisons will be contaminated by target inconsistency rather than model quality. [7]

Data sources and vintage-safe dataset design

Source assessment

The table below focuses on what matters operationally: legal pre-noon availability, history, whether point-in-time vintages can be reconstructed, and the main implementation traps.

Source	Access method	What it gives you	History / frequency / granularity	Vintage availability at 12:00 CET	Main gotchas
Elering	Public dashboard and Swagger-documented API	EE/LV/LT/FI prices, system data, some load / generation views	Public dashboard/ API is clearly available; indexed docs do not expose a clear public retention SLA I could verify. [8]	**Unclear** in indexed public docs	Good operational source for Estonia-specific current data, but official indexed documentation is weak on retention, latency SLAs, and forecast-vintage reconstruction. Treat as useful operational feed, not as your sole historical truth store. [9]

Source	Access method	What it gives you	History / frequency / granularity	Vintage availability at 12:00 CET	Main gotchas
Nord Pool	Data Portal, customer APIs, paid market data services	Day-ahead prices, capacities, intraday, market data products	Official market-data services advertise history back to 1992; APIs exist for customers; public portal exists but broad historic programmatic access is more limited than the commercial offer. [10]	For settled prices, vintage is not the issue; for capacities and market data, use the publication timestamp	Best authoritative source for market results and a key source for target values. Beware that after 30 September 2025, native DA prices are quarter-hourly in SDAC. [11]

Source	Access method	What it gives you	History / frequency / granularity	Vintage availability at 12:00 CET	Main gotchas
ENTSO-E Transparency Platform	Web API and platform downloads	Pan-European load forecasts, generation forecasts, flows, outages, REMIT-linked publications	Official platform launched in 2015; at least five years of detailed history are retained for core transparency items. [12]	**Partial but operationally tricky**	This is the richest open public fundamentals source, but revisions and corrections are part of the design, and independent reviews document usability and data-quality shortcomings. In practice, you should archive your own pulls or at least preserve publication / document timestamps instead of trusting a later backfill as the original noon vintage. [13]

Source	Access method	What it gives you	History / frequency / granularity	Vintage availability at 12:00 CET	Main gotchas
ECMWF	Open data, dissemination feeds, or licensed archive	HRES / ENS weather forecasts, including wind / irradiance surrogates	HRES / IFS runs 4 times daily at 00/06/12/18 UTC; data are released according to dissemination schedule. [14]	**Yes, if you archive model-run-based data**	Excellent for point-in-time vintage if you work by model run, but heavier operationally than lightweight APIs. For a noon bid, the relevant usable run is usually 00 UTC or earlier dissemination available before 12:00 CET. [15]
DWD ICON	Open Data server / DWD products	ICON-EU and related regional forecasts	ICON-EU offers four main runs at 00/06/12/18 UTC and additional 3-hourly runs; hourly output is available for the short horizon. [16]	**Yes, if stored by run**	Strong open alternative for Baltic/Nordic weather features; easier than ECMWF licensing, but you still need a run-aware archive for vintage-safe backtests. [16]
Open-Meteo	Historical Forecast API / Previous Runs API	Archived forecast weather products from multiple underlying models	Past forecast archives are available from 2022 onward and are updated daily; previous run access is exposed in the API docs. [17]	**Yes, and unusually convenient**	This is the fastest way to build a vintage-safe prototype. The tradeoff is less control and less direct transparency than raw ECMWF / DWD ingestion. [18]

Source	Access method	What it gives you	History / frequency / granularity	Vintage availability at 12:00 CET	Main gotchas
ICE / EEX commodity data	Exchange pages, vendor feeds, or market-data subscriptions	TTF gas, API2 coal, Brent, EUA futures / settlements	Exchange-traded products with daily settlement / intraday data; exact history access depends on license or vendor route. [19]	**Yes for lagged settlements; same-day intraday depends on feed**	Use lagged D-1 settlement curves for clean vintage handling. For this problem, commodity features are slower-moving state variables, not high-frequency noon-edge alpha. [20]
Hydrology	NVE / Statnett and commercial Nordic hydrology sources	Reservoir filling and related hydro state	Statnett documents weekly reservoir data by spot area with history back to 1993. [21]	**Yes, at weekly publication cadence**	Important for Nordic price formation, but low frequency. Treat as weekly state, not an hourly driver. [21]
JAO	Publication Tool and APIs, including Nordic CCR views	Cross-zonal capacity publications and related constraint data	Public tool and APIs are explicitly exposed. [22]	**Usually yes if archived at publication time**	Important gotcha: the Nordic handbook notes that `dateTimeUtc` in responses actually follows CET-style timestamps, so timezone validation is mandatory. [23]

Source	Access method	What it gives you	History / frequency / granularity	Vintage availability at 12:00 CET	Main gotchas
Transmission / outages	Nord Pool REMIT UMM, Elering, Fingrid operational messages	EstLink 1 / 2 outages, restoration timing, transmission unavailability	Real-time and archival message streams exist. [24]	**Yes, if you persist messages as published**	EstLink outages have been materially important since late 2024. Message timestamps and estimated restoration times change, so archive the message stream itself, not only final status. [25]
Calendar	Official Estonian law / state portals	Public holidays and special calendar effects	Official statute is stable and explicit. [26]	Not a vintage issue	Use Estonia-specific holidays, not generic European holiday calendars. Midsummer and restoration / independence effects matter more than a generic “EU holiday” flag. [27]

Recommended dataset structure

The **canonical raw store** should be event-based and vintage-aware: one row per `(source, publication_timestamp, valid_from, zone, variable)` entry, with the exact value visible before or at 12:00 CET on day D. This raw store should preserve the native market-time timestamps from the source. That raw layer is not only for auditability; it is the only way to guarantee that backtests do not consume later revisions from ENTSO-E, updated interconnector statuses, or weather runs that were not yet disseminated. This is the highest-value engineering choice in the whole project. [28]

For **modeling**, I recommend two derived views:

First, a **daily wide table** for the primary architecture, with one row per forecast origin day D and a 24-dimensional target for D+1. This is the cleanest format for LEAR-style multi-output models, daily profile learning, and coherent scenario generation. It naturally represents the object you actually optimize over: the full next-day price path.

Second, an **hourly long table** with one row per `(forecast origin day D, delivery hour h on D+1)` for tree-based fallback models. This format works well with gradient boosting using `hour`, `day-type`, and interacted exogenous features.

The raw store should remain the source of truth; these views are for model convenience. This recommendation is a reasoned inference from the model classes and the 24-dimensional EPF literature, rather than a formal standard. It is especially attractive under thin data because it avoids proliferating separate data pipelines for each architecture. [29]

Target definition

For the initial product, define the target as **EE day-ahead hourly decision prices in EUR/MWh** over the delivery day D+1, with timestamps stored in **market time (CET/CEST)** and only localized to Europe/Tallinn for display. If you continue to insist on a 24-element output after the 15-minute SDAC go-live, then the aggregation rule must be written into the schema itself, for example as the hourly Nord Pool index or a fixed arithmetic mean of the four quarter-hours. Do not mix native pre-September-2025 hourly prices with post-September-2025 ad hoc hourly averages without a formal target-definition note. [30]

DST should be handled in the raw store by native market periods. At the modeling layer, either use variable-length delivery-day objects on DST transitions, or map to an explicit 24-hour operational layer only after a documented aggregation rule. Hiding DST by naive local-time resampling is a quiet but serious source of label noise in electricity markets. [31]

Feature set

The first model should use six feature families.

Lagged target prices. Include EE own-price lags at 24 h, 48 h, 168 h, and 336 h; include lagged neighboring-zone prices for Finland [32], Latvia [33], and Lithuania [34] at the same lag structure; and include selected cross-hour lags from the previous day and previous week to capture intraday dependence. This is standard, well-established EPF practice and sits at the heart of LEAR / lasso-style high-dimensional models. [35]

Exogenous forecasts aligned to delivery hour. Use day-ahead load forecasts, wind and solar generation forecasts, and cross-border flow / capacity variables for EE and the most coupled neighbors. For Estonia specifically, a pre-noon information set should prefer ex ante forecasts from Elering / ENTSO-E / weather models, not realized load or realized renewable generation. The important variables are not merely wind and solar separately, but the net power-balance state faced by the Baltic-Finnish interface. [36]

Capacity and outage features. Include EstLink 1 and EstLink 2 available capacity, binary outage flags, hours-to-restoration proxies, Baltic border capacities, and JAO / TSO-published cross-zonal limits as soon as they are published. In this corridor, transmission conditions are too important to leave implicit. [37]

Calendar and holiday features. Use day-of-week, month, sinusoidal annual encodings, bridge-day flags, and explicit Estonian public-holiday indicators. A generic EU holiday flag is too crude for a market with strong local social-load patterns and summer holiday effects. [26]

Weather and physics-derived features. Include temperature, wind speed, cloud cover / irradiance, daylight duration, solar elevation proxy, and derived residual-load features such as `load_forecast - wind_forecast - solar_forecast`, plus intraday ramp variables for load and renewables. For Estonia's latitude and seasonal day-length swing, those physics-based variables are more transportable than bare hour-of-day dummies. This is partly well-established and partly reasoned inference tailored to the Baltic setting. [38]

Slow-moving regime features. Add lagged TTF gas, EUA, Brent, Nordic reservoir levels, and possibly front-month / front-quarter spreads as daily state variables. In a one-day-ahead Baltic model these will rarely drive today-versus-tomorrow micro-movements by themselves, but they help encode the broader cost environment and scarcity regime. This is best treated as a modest prior-state block, not a high-importance feature family. [39]

Vintage rule, lag structure, and missing-data policy

The **vintage rule** should be explicit and ruthless: every feature in a training row for forecast origin D must equal the value that was visible **at or before 12:00 CET on D**. If a source is revised later, the later revision is irrelevant for that row. The practical implication is that ENTSO-E, weather forecasts, outage messages, and capacity publications must be stored with their publication time or model-run time and filtered accordingly in both training and inference. This is the single most important control against leakage. [40]

The lag structure should start with **24 h, 48 h, 168 h, and 336 h**, which is standard in EPF because it captures yesterday, two days ago, same weekday last week, and the two-week memory that often stabilizes weekly periodicity without bloating the design matrix. In a pooled LEAR-type model, I would also add selected cross-hour lags from the previous day and previous week; in a long-format boosting model, those become explicit engineered columns such as `ee_price_d_1_h_same`, `ee_price_d_7_h_same`, `fi_price_d_1_h_same`, and `ee_prev_evening_peak_mean`. [41]

For missing data, use a hierarchy rather than a one-size-fits-all imputer. Short gaps in lagged prices can be forward-filled inside the raw market data only when the exchange itself reports continuity; weather gaps should be replaced by same-run neighboring-grid interpolation if available; forecast fundamentals should prefer same-source fallback and then be flagged with explicit "imputed" indicators. Missingness itself is informative in outage-heavy systems, so do not silently smooth it away. This recommendation is mostly reasoned inference from operational forecasting practice.

Auxiliary datasets for transfer and pooling

If you pursue transfer learning or multi-task modeling, align auxiliary data from Finland, Latvia, Lithuania, Sweden SE3, and optionally Germany-Luxembourg into the same schema with a `zone_id`, market-time timestamps, and features normalized by local scale. Use local z-scoring or robust scaling and include zone embeddings or zone identifiers rather than attempting to force raw price levels into one pooled model without a market label. This is the safest way to borrow shape information while allowing level differences to remain market-specific. Transfer-learning work in EPF finds that pretraining on richer markets and fine-tuning on the target can significantly improve accuracy, but only if the architecture remains disciplined about market identity and rolling evaluation. [42]

Candidate architectures

The comparison below synthesizes the established EPF literature on reduced-form models, lasso / LEAR and multivariate structures, benchmark studies, NBEATSx, TFT, transfer learning, and the newer time-series foundation-model papers. The ratings are my synthesis for **this exact Estonia-2025/26 thin-data setting**, not universal rankings. [43]

Architecture	Thin-data fit	Interpretability	Train / inference cost	Probabilistic capability	Assessment for this problem
Naive baselines: yesterday / last week / similar-day	Very high	Very high	Minimal	Weak	Mandatory baselines. They will be hard to beat during stable weeks and are essential for significance tests.
LEAR, per-hour	High	High	Low	Moderate via residual bootstrap / quantile post-processing	Strong candidate. Very data-efficient, well-understood, and robust under daily recalibration. Best as a core expert.
LEAR, pooled / multivariate daily output	High	High	Low to medium	Moderate	Slightly better fit to the full 24-hour object and cross-hour dependence. Often preferable here to 24 entirely separate local models.
MSTL decomposition plus per-component models	Medium to high	Medium to high	Medium	Moderate	Useful as a feature-engineering layer or residual stabilizer, but not my primary standalone architecture under 12 months.

Architecture	Thin-data fit	Interpretability	Train / inference cost	Probabilistic capability	Assessment for this problem
LightGBM / XGBoost with hour-as-feature	High	Medium	Low to medium	High for marginal quantiles	Excellent fallback / companion expert. Strong under short-window training, good with nonlinear interactions, but needs a coherence layer for full-day scenarios.
Shared-trunk DNN with 24 heads	Medium	Low to medium	Medium	Medium to high	Viable only with transfer / multi-market pooling. Too fragile as a purely local 12-month model.
NBEATSx	Medium	Medium	Medium to high	Medium to high	Strong literature pedigree in EPF, but local-only Estonia data are too thin. Becomes interesting after pretraining or multi-market transfer.
TFT	Low to medium	Medium	High	High	Architecturally attractive because it handles known-future covariates and quantiles, but it is too data-hungry and tuning-sensitive for a first production model here.

Architecture	Thin-data fit	Interpretability	Train / inference cost	Probabilistic capability	Assessment for this problem
Chronos-Bolt zero-shot	Medium	Low	Low inference, no training	High	Good benchmark and ensemble member. Useful for cold-start sanity checks, but weak as a full covariate-aware production forecaster.
Chronos-Bolt fine-tuned	Medium	Low	Medium	High	Better than zero-shot, but still less natural than purpose-built EPF models for rich exogenous inputs in this market design.
TimesFM zero-shot / few-shot	Medium	Low	Medium	Historically weaker probabilistic support in common open releases	Useful benchmark, especially when only a raw univariate series is available. Not my preferred core architecture for covariate-heavy DA trading.
Time-MoE zero-shot / few-shot	Medium	Low	Medium to high	Less turnkey than Chronos-Bolt for this use case	Strong general TSFM, but overpowered and operationally heavier than needed for a 24-hour DA Estonia model.
Merit-order / curve forecasting	Medium	High in mechanism terms	High data and engineering cost	High	The best “research upside” model if bid curves and capacities are available with proper vintages. Not the first model to ship.

Architecture	Thin-data fit	Interpretability	Train / inference cost	Probabilistic capability	Assessment for this problem
Ensemble combinations	Very high	Medium to high	Medium	High	Best overall answer. Combining a regularized statistical core with a nonlinear expert is the most robust design under thin data.

Architecture-by-architecture judgment

Naive baselines. Well-established. Keep, report, and test them seriously. In volatile but strongly seasonal markets, last-week-same-hour and similar-day baselines often remain surprisingly competitive, and the EPF literature repeatedly criticizes studies that skip them. These are not toy models here; they are the minimum bar for claiming value. [44]

LEAR. Well-established and especially attractive here. Lasso-based high-dimensional ARX models were developed precisely to capture the strong cross-hour dependence structure of day-ahead prices while remaining regularized enough for daily rolling estimation. The EPFTOOLBOX documentation explicitly treats LEAR as one of the state-of-the-art benchmark models, and the lasso / high-dimensional literature repeatedly shows that simple, recalibrated regularized models remain extremely hard to beat. In this Estonia setting, LEAR's main virtues are not just accuracy but *stability*, *transparency*, and *compatibility with lagged neighboring-zone features and published forecasts*. [45]

Between per-hour and pooled variants, my preference is **a pooled multivariate daily-output LEAR-style formulation** for the primary implementation, with the option to keep per-hour local experts in the ensemble. The univariate-vs-multivariate paper by Florian Ziel and Rafał Weron does not show a universal multivariate win, but it does show that the multivariate edge can be real and, more importantly here, that combining both approaches is often better than choosing one dogmatically. Under one year of data, that argues for a small diversified family of lasso experts rather than a single monolithic specification. [46]

MSTL decomposition plus per-component models. Mostly reasoned inference with some supporting decomposition literature. MSTL is attractive because the price path contains multiple seasonalities, but decomposition does not create information and can easily become another tuning surface in a tiny sample. I would not deploy it as the lead architecture. I *would* use it experimentally as a preprocessing step or feature generator, especially if it sharpens weekly and intraday structure before a simpler model. [47]

Gradient boosting. Strong candidate, especially as an ensemble member. Recent short-window studies report LightGBM as particularly robust when training windows are only 45–60 days, and boosting is naturally good at nonlinear interactions among load, renewables, outages, temperature, and hour / holiday effects. In this problem, LightGBM with `hour-as-feature`, interaction features, and rolling retraining is the best non-neural fallback. Its main weakness is that independent quantiles and point forecasts are not automatically coherent across the full 24-hour path. That can be fixed, but the coherence layer is separate. [48]

Shared-trunk DNN, NBEATSx, and TFT. These are the wrong first answer to the problem as stated, unless you combine them with aggressive transfer learning or multi-market pretraining. NBEATSx has excellent EPF credentials and delivers interpretable decompositions; TFT is explicitly built for mixed static, historical, and known-future covariates with multi-horizon quantiles; and the Europe-focused DNN literature shows that connected-market features can help. But all three have a serious small-sample problem in purely local Estonia-2025/26 training. Their edge emerges when more data or transfer structure exist. Without that, they consume the modeling budget on tuning variance rather than signal extraction. [49]

Foundation models. My view is measured rather than dismissive. TimesFM, Chronos, Chronos-Bolt, and Time-MoE all demonstrate that zero-shot or few-shot forecasting can be surprisingly competitive, especially when local data are scarce. Chronos-Bolt is particularly relevant because it is fast, memory-efficient, and natively produces multi-step quantile forecasts. But this family still has two disadvantages for this specific DA electricity problem: first, the information edge is mostly in *rich structured covariates* and market timing, not in the univariate price series alone; second, the operational question is not just “forecast a series,” but “forecast a leakage-safe, pre-noon, exogenous-aware 24-hour market object.” That keeps foundation models in the role of benchmark, fallback benchmark, or ensemble member rather than the main production brain. [50]

Merit-order / curve forecasting. This is the most intellectually attractive alternative and the highest-upside research track. The Ziel-Steinert X-model and the newer curve-based work by Guillaume Koechlin and coauthors show why: if you forecast the supply and demand curves or their functional representations, you model the market mechanism itself, and you can derive both prices and distributions more naturally. For Estonia, where interconnector constraints and net imports matter materially, that mechanism-first logic is compelling. The problem is not theory; the problem is engineering and data. You need consistent historical curves or curve proxies, careful capacity and outage vintages, and likely more infrastructure than the first rollout deserves. My recommendation is to treat this as a phase-two or phase-three research line, especially if access to curves becomes available. [51]

Ensembles. This is the category I would actually choose. The literature already suggests gains from combining univariate and multivariate models, and current practice in volatile markets increasingly favors forecast combination because no single architecture dominates under all regimes. Under 12 months of in-regime data, ensembling is not a hedge against indecision; it is the rational response to model-class uncertainty. [52]

Strategies for the 12-month constraint

The thin-data problem cannot be solved by architecture alone. Some strategies genuinely add information; others merely make overfitting less destructive.

The ranking below is my recommendation for this project.

Strategy	Rank	Does it truly help the thin-data problem?	Judgment
Daily recalibration with sliding window	High	**Partly**	Necessary but not sufficient. It adapts to drift; it does not create information.
Transfer learning from richer neighboring markets	High	**Yes**	One of the few strategies that truly adds transferable information if market identity is preserved.
Multi-task learning across Baltic / Nordic zones with zone embedding	High	**Yes**	Strong candidate when auxiliary data are available and the architecture is kept small.
Hybrid statistical + fundamental features	High	**Yes**	Encodes mechanism and reduces reliance on pure pattern repetition.
Bayesian / heavily regularized models with priors	Medium to high	**Partly**	Very useful to stabilize learning, but they still need informative priors or pooled data.
Foundation-model zero-shot / few-shot	Medium	**Partly**	Solves the data-scarcity problem for raw series patterns, but not fully for market-specific covariate-aware forecasting.
Data augmentation	Low	**Usually no**	Mostly papers over the problem unless the augmentation is mechanism-aware and validated very carefully.

What actually solves the problem

Transfer learning and multi-market pooling are the most credible solutions. Transfer-learning evidence in EPF shows that pretraining on source markets and fine-tuning on the target can significantly improve performance. The key caveat is that you should transfer *representations* and *shape information*, not assume that Estonia is merely a scaled copy of Finland or Germany-Luxembourg. Baltic and Nordic pooling is the most plausible first step because the interconnections and hydro / wind / import structure create some genuine commonality. [42]

Hybrid statistical-plus-fundamental modeling is the other real solution. This does not "borrow more data" from elsewhere, but it reduces the sample burden by encoding causal structure: load, wind, solar, capacity, outages, residual load, and hydro state. In a market with a real regime break, that matters more than squeezing another few basis points from a deeper black box. This is the strongest argument for keeping LEAR / boosting models at the center rather than moving immediately to TFT or fully local NBEATSx. [53]

What mainly reduces damage

Daily recalibration with a sliding window is indispensable, and the calibration-window literature shows exactly why: selecting one fixed window is risky because the best window length varies across markets and periods. But recalibration should be understood correctly. It is an adaptation device, not a substitute for missing information. It reduces stale-pattern risk; it does not supply the seasonal or structural diversity that a one-year sample lacks. In this project, I would use daily recalibration everywhere, but I would not pretend that it solves the scarcity problem on its own. [54]

Bayesian or strongly regularized models also sit in this category. They can be extremely helpful, especially for pooled LEAR variants, shrinkage on cross-market coefficients, or hierarchical priors on zone effects. But unless the priors are informed by closely related source markets, they mostly stabilize rather than enrich. My recommendation is therefore to use heavy regularization first and full Bayesian machinery only if the team already has that capability.

What is useful but secondary

Foundation-model zero-shot and few-shot methods are useful, but not in the way hype often suggests. They are a great way to get a fast benchmark that can be surprisingly strong with almost no local training. They are also attractive if you temporarily lack clean neighbor-market covariates. But for this specific noon-before-gate-closure problem, the edge is not primarily in generic sequence continuation. It is in leakage-safe integration of known-future fundamentals and transmission conditions. That is why I rank them below transfer and hybrid statistical-fundamental approaches. [55]

Data augmentation is the most likely way to fool yourself. Synthetic price paths, bootstrapped days, or generic perturbations may make neural models look happier during training, but unless the augmentation respects market coupling, weather-capacity consistency, and spike mechanics, it is usually a cosmetic fix. I would avoid it in the first two rollout phases.

Seasonal drift, regime modeling, and output design

Seasonal and structural regime drift

The problem statement is exactly right to emphasize that price regimes shift in clock-time over the year. Estonia's winter mornings, shoulder-season ramps, and the moving location of the solar-driven price dip are not just different levels of the same daily curve; they are different mappings from weather and system state to price. A pure clock-time encoding is therefore too weak. It helps the model memorize fixed intraday seasonality, but it cannot explain why the daytime low moves with daylight or why evening scarcity becomes steeper when imports are constrained. This is where physics-based features earn their keep. [56]

My recommendation is to model regime drift with **soft regime descriptors**, not separate hard regime models. In practice that means combining `hour`, `day-of-year`, and holiday features with daylight duration, solar-angle proxies, residual load, renewable ramps, and transmission / outage flags. This lets the model discover that "midday" is a seasonal phenomenon rather than a fixed clock slot. It is the most data-efficient way to encode the drift without fragmenting a one-year sample into tiny seasonal buckets. This is a reasoned inference, but it is strongly supported by the success of exogenous-aware EPF models and residual-load-driven probabilistic work. [57]

Could daily recalibration substitute for explicit regime modeling? Only partly. Rolling recalibration will keep coefficients closer to the current regime, but it cannot infer an unseen summer-to-winter migration of the shape if the features do not tell it *why* the regime moved. In other words, recalibration is a local correction mechanism; physics-based features are the transport mechanism across seasons. With only one post-break year, you need both. [54]

Twenty-four separate models versus multi-output versus sequence models

For this use case, I would not choose 24 completely separate production models as the final design, even though local per-hour models remain a valuable benchmark. Separate models throw away cross-hour structure exactly where a battery or solar-plus-storage operator cares most: the joint profile and the width of peaks and troughs. They can still be useful expert components in an ensemble, especially for spike-prone hours, but they should not be the only final interface. [58]

I prefer a **multi-output daily model** as the production abstraction. That does not require a deep sequence model. A pooled LEAR, a daily vector regression, or an ensemble whose final layer outputs a coherent daily path is enough. The point is to keep the forecast object aligned with the decision object. This is one of the strongest arguments for a daily wide-table design. It also future-proofs the pipeline if you later upgrade from 24 hourly outputs to 96 quarter-hour outputs. This recommendation is a reasoned inference grounded in the multivariate-vs-univariate literature and the downstream optimization need. [59]

Sequence models such as TFT are most attractive when you have abundant data, rich known-future covariates, and you want native multi-horizon quantiles from one model. That is all true in principle here, but the data volume is wrong. So I would keep “sequence model” as a research direction, not as today’s default.

Point forecasts versus probabilistic forecasts

For the user’s downstream application, **probabilistic forecasts are not optional**. A small solar-plus-battery operator does not merely want the expected price path; they need a view on uncertainty to size bids and manage the cost of over- and under-commitment. Recent work on EPF and battery trading makes the broader point clearly: quantiles are useful, but the real object of value is a coherent stochastic representation of the whole day, not 24 unrelated marginal intervals. [60]

For that reason, the right production output is:

1. a **point forecast** for the 24-hour path,
2. a **small set of calibrated marginal quantiles** such as P10 / P50 / P90, and
3. a **scenario set** for the full day, ideally 100–500 paths preserving cross-hour dependence.

The simplest way to get there in phase one is not a heavy generative model. It is a **point ensemble plus residual-based probabilistic layer**: estimate the marginal uncertainty by quantile models or conformalized residuals, then restore cross-hour dependence using a shrinkage Gaussian copula, empirical residual bootstrap, or day-level residual resampling from similar regimes. Only later, if the dataset and evaluation support it, would I move to a dedicated scenario generator such as a normalizing flow. [61]

Evaluation methodology

The evaluation design should be conservative because the sample is small enough that a careless split can make almost any model look good.

Recommended walk-forward protocol

Use a **strict rolling-origin backtest** with daily forecast origins and full vintage reconstruction. Start only when all required lag features are available. For each origin day D in the test span, fit or recalibrate the model using data available up to 12:00 CET on D, forecast delivery day D+1, score the full next-day path, and move forward one day. This matches the operational forecast loop and respects both recalibration and vintage constraints. [62]

Given only about 12 months of usable data, I recommend a **single final holdout block of the most recent 2–3 months** and an earlier walk-forward development period for model selection. Concretely, an illustrative split would be:

- earliest in-regime period to month 8 or 9: development via rolling-origin backtests,
- month 10 to month 12: frozen final evaluation.

Inside development, compare calibration windows such as 42, 56, 84, and 112 days for LEAR / LightGBM families rather than assuming a single universal window. This is far more defensible than spending the limited data on a large random hyperparameter search. [63]

Metrics

For **point forecasts**, report **MAE** and **RMSE** as primary metrics, and optionally **sMAPE** only as a secondary descriptive metric. Do **not** use ordinary MAPE as a headline metric because electricity prices can be zero, near zero, or negative, and MAPE becomes unstable or misleading in precisely those conditions. That caveat is longstanding in EPF. [64]

For **probabilistic forecasts**, report **pinball loss** for the selected quantiles, **CRPS** where available, and a multivariate score such as the **energy score** for full-day scenarios. The marginal quantile metrics capture calibration and sharpness hour by hour; the multivariate score penalizes incoherent scenario generation, which matters for battery scheduling. [65]

Reporting granularity

Report results in four layers:

- overall aggregated test-sample score,
- per-hour average score,
- regime slice score such as overnight / morning ramp / daytime / evening peak,
- spike-hour subset score.

The spike subset should be predeclared, for example as top-decile absolute prices, top-decile residuals over a rolling median, or hours exceeding a dynamic regime-dependent threshold. A model that wins on average but fails on scarcity hours may be unacceptable for trading. This is a reasoned inference, but it is fully consistent with recent spike-focused and probabilistic EPF work. [66]

Statistical tests

Use **Diebold-Mariano** tests for pairwise forecast comparison on daily loss differentials, and **Giacomini-White** tests when you want to assess conditional predictive ability under rolling estimation. In this setting, the relevant comparison unit is the **whole day**, not 24 independent hourly tests only, because the decision problem is daily and the residuals are cross-hour dependent. Hour-level DM tests can still be useful as diagnostics, but the governance decision should be based on daily multivariate loss. [67]

Validation criterion for architecture choice

A model should become the production champion only if it clears all three gates:

- it beats the naive baselines and the LEAR core in out-of-sample point accuracy,
- the improvement is statistically significant on the daily loss differential, and
- its probabilistic outputs improve pinball / CRPS / energy score without degrading operational plausibility.

That standard sounds obvious, but it is exactly where many EPF studies become unreliable: too-short test spans, weak baselines, and no significance testing. [44]

Final recommendation

Recommended primary architecture

My recommended primary architecture is a **hybrid daily-recalibrated ensemble** with three layers:

Core expert: a **pooled LEAR-style ARX model** on the daily wide table, using lagged EE and neighboring-zone prices, pre-noon exogenous forecasts, transmission / outage variables, calendar effects, and physics-based regime features.

Nonlinear expert: a **short-window LightGBM quantile model** on the long hourly table with `hour-as-feature`, residual-load variables, outage / capacity flags, weather, and interactions.

Probabilistic layer: a **scenario generator built from the ensemble residual structure**, initially via calibrated marginal quantiles plus a dependence-restoring copula / day-level bootstrap rather than a heavyweight generative deep model.

This recommendation is partly well-established and partly reasoned synthesis. The well-established part is that regularized ARX / LEAR models are top-tier EPF baselines, short-window boosting is strong in data-constrained volatile markets, and forecast combination is often beneficial. The synthesis is that Estonia-2025/26 needs *all three* together because no single architecture simultaneously gives thin-data robustness, nonlinear adaptation, and full-day probabilistic usability. [68]

Why this, and not TFT / NBEATSx / foundation models as the lead? Because the scarce resource here is not GPU time; it is trustworthy in-regime information. The hybrid ensemble makes the best use of that scarce information while staying auditable and easy to recalibrate daily. It also de-risks deployment: if the nonlinear expert degrades during a new outage regime, the LEAR core remains a strong anchor.

Recommended fallback

The fallback model should be a **pure LEAR family**: preferably two closely related lasso experts, one slightly shorter-window and one slightly longer-window, averaged or adaptively combined, with asinh-style variance stabilization if needed for volatile periods and residual bootstrapping for uncertainty bands. If engineering time is short, this is the first thing I would ship. It is transparent, fast, and extremely defensible. [69]

Architectures I would not prioritize first

I would not make **TFT** or **local-only NBEATSx** the first production model. I would benchmark them only after the data pipeline, LEAR, and LightGBM are in place, and ideally only with pooled multi-market training.

I would not make **Chronos-Bolt, TimesFM, or Time-MoE** the production champion on day one. I would absolutely benchmark them, and I would consider Chronos-Bolt as an ensemble member because it is fast and probabilistic. But I would not ask a mostly univariate foundation model to carry the full burden of a vintage-safe, exogenous-aware Baltic day-ahead market model. [70]

I would not make **merit-order curve forecasting** the first deployment target unless historical bid curves and clean publication vintages are already available internally. If they are, the answer changes materially, because curve models become much more attractive. [71]

Phased rollout plan

First phase. Build the vintage-safe raw data store and two benchmark families only: naive baselines and LEAR, plus the LightGBM hourly fallback. Freeze target definition. Get the walk-forward backtest right. This is the phase where most project risk is eliminated.

Second phase. Add probabilistic outputs and scenarios. Start with quantile LightGBM or conformalized residuals and a dependence-restoring bootstrap / copula. Run battery-oriented stress tests, but do not let trading PnL alone override statistical validation. [72]

Third phase. Add multi-market transfer / pooling, preferably with Finland, Latvia, Lithuania, and selected Nordic zones. Re-evaluate whether a compact shared-trunk DNN or NBEATSx becomes competitive once the auxiliary data are aligned.

Fourth phase. Run research tracks on foundation-model ensembling and merit-order / curve forecasting. Only promote them if they beat the hybrid baseline on both daily statistical metrics and scenario quality.

Highest-leverage decisions

The two or three highest-leverage decisions are:

Target definition after the 15-minute transition. If the true operational task is now 96 quarter-hours, then a 24-hour model is already an aggregation model, not a native market model. That changes the whole recommendation frontier. [11]

Vintage-safe data engineering. If ENTSO-E, weather, outages, and capacities are not archived as of noon, any sophisticated backtest can become invalid. This is a bigger source of error than most architecture choices. [73]

Whether to pool across markets. A carefully designed Baltic / Nordic transfer setup could move the needle materially; a purely local model probably leaves performance on the table. [74]

Dominant risks

The dominant risks are not mainly algorithmic.

The first is **silent leakage through revised exogenous series**. The second is **overconfidence from one seasonal cycle**, especially around spring and autumn ramp shifts. The third is **producing incoherent probabilistic outputs** by modeling marginal quantiles independently and then using them inside a battery optimizer as if they were a joint distribution. Those risks are operationally larger than "choosing the wrong deep-learning architecture." [75]

Executive summary

The best architecture for this task is not a single flashy model. It is a **hybrid ensemble built around a daily-recalibrated, heavily regularized LEAR-style core, complemented by a short-window LightGBM expert and a separate probabilistic scenario layer**. That combination is the best fit to the actual constraints: simultaneous SDAC clearing, strict noon information availability, one year of in-regime data, a February 2025 synchronization break, and the post-September 2025 15-minute target-definition issue. The evidence base for the ingredients is strong: lasso / LEAR remains one of the most reliable classes in EPF; short-window boosting performs well under constrained data; and combining experts is usually safer than betting on one class in volatile markets. [76]

The recommended data stack is equally important. For market results, use Nord Pool as the authoritative target source. For Estonia-specific operational feeds, use Elering where convenient, but do not rely on it as your only historical truth store because the indexed public docs do not clearly expose retention and vintage guarantees. For exogenous fundamentals, ENTSO-E is the richest public source, but you must archive it by publication time because revisions are part of the platform design. For weather, the quickest vintage-safe prototype path is Open-Meteo's Historical Forecast / Previous Runs APIs; for production-grade control, move to raw ECMWF or DWD run-based archives. Add JAO capacities, EstLink outage messages, weekly Nordic hydrology, lagged commodity settlements, and explicit Estonian holidays. [77]

The recommended dataset schema has two layers. The **canonical raw layer** should store native, point-in-time source values with publication timestamps. The **model layer** should expose a daily wide table for the primary 24-output architecture and an hourly long table for the boosting fallback. The feature set should include own-price and neighbor-price lags, known-future load / wind / solar forecasts, transmission and outage variables, calendar features, weather variables, and derived physics features such as residual load and renewable ramps. Every training row must reflect the information state visible by **12:00 CET on day D**. [78]

The top three risks are: **feature-vintage leakage**, **target-definition ambiguity after the 15-minute SDAC transition**, and **probabilistic incoherence when downstream optimization needs full-day scenarios rather than independent hourly quantiles**. Those are the failure modes most likely to create misleading backtests or poor trading performance. [79]

The two questions that would most change the recommendation are these. First, **is the real operational target 24 hourly decision prices, or should the system forecast the native 96 quarter-hour prices and aggregate only afterward?** If the answer is "96 quarter-hours," sequence and curve-based architectures become more attractive. Second, **can you build or obtain a genuine noon-vintage archive for ENTSO-E forecasts, weather runs, capacities, outages, and possibly bid curves?** If yes, the merit-order / curve-forecasting line becomes much more compelling; if no, the hybrid LEAR-plus-LightGBM ensemble becomes even more clearly the right answer.

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